

# DS\_A1\_Q5

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## 1 Q5

To show that test MSE of model  $Y = f(x) + \epsilon$ ,

$$\mathbb{E}[(Y_0 - \hat{f}(x_0))^2] = \text{var}(\hat{f}(x_0)) + b_f(\hat{f}(x_0))^2 + \text{var}(\epsilon)$$

We could write the expression of test MSE first,

$$\begin{aligned}\mathbb{E}[(Y_0 - \hat{f}(x_0))^2] &= \mathbb{E}[(f(x_0) + \epsilon - \hat{f}(x_0))^2] \\ &= \mathbb{E}[(f(x_0) - \hat{f}(x_0) + \epsilon)^2] \\ &= \mathbb{E}[(f(x_0) - \hat{f}(x_0))^2] + 2\mathbb{E}[(f(x_0) - \hat{f}(x_0))\mathbb{E}[\epsilon] + \mathbb{E}[\epsilon^2]]\end{aligned}$$

Since  $\epsilon$  is a random error term that is independent of  $x$  and has mean zero, so

$$\mathbb{E}[\epsilon] = 0, \quad \text{var}(\epsilon) = \mathbb{E}[\epsilon^2]$$

Substitute the formulas into the derived test MSE expression,

$$\begin{aligned}\mathbb{E}[(Y_0 - \hat{f}(x_0))^2] &= \mathbb{E}[(f(x_0) - \hat{f}(x_0))^2] + 2\mathbb{E}[(f(x_0) - \hat{f}(x_0)) \cdot 0] + \text{var}(\epsilon) \\ &= \mathbb{E}[(f(x_0) - \hat{f}(x_0))^2] + \text{var}(\epsilon)\end{aligned}$$

To further prove the result, we transform  $f(x_0) - \hat{f}(x_0)$  as follows.

$$f(x_0) - \hat{f}(x_0) = (f(x_0) - \mathbb{E}[\hat{f}(x_0)]) + (\mathbb{E}[\hat{f}(x_0)] - \hat{f}(x_0))$$

Substitute to  $\mathbb{E}[(f(x_0) - \hat{f}(x_0))^2]$ , we can obtain that,

$$\mathbb{E}[(f(x_0) - \hat{f}(x_0))^2] = \mathbb{E}[(\mathbb{E}[\hat{f}(x_0)] - \hat{f}(x_0))^2] + \mathbb{E}[(f(x_0) - \mathbb{E}[\hat{f}(x_0)])^2] + 2\mathbb{E}[(\mathbb{E}[\hat{f}(x_0)] - \hat{f}(x_0))(f(x_0) - \mathbb{E}[\hat{f}(x_0)])]$$

Once more,  $\epsilon$  is a random error term that is independent of  $x$  and has mean zero, so

$$\mathbb{E}[\hat{f}(x_0)] = \mathbb{E}[f(x_0) + \epsilon] = \mathbb{E}[f(x_0)] + \mathbb{E}[\epsilon] = f(x_0)$$

So we can infer that  $2\mathbb{E}[(\mathbb{E}[\hat{f}(x_0)] - \hat{f}(x_0))(f(x_0) - \mathbb{E}[\hat{f}(x_0)])] = 0$ .

According to the property of variance and the definition of bias,

$$\begin{aligned}\text{var}(\hat{f}(x_0)) &= \mathbb{E}[(\mathbb{E}[\hat{f}(x_0)] - \hat{f}(x_0))^2], \\ b_f(\hat{f}(x_0))^2 &= (f(x_0) - \mathbb{E}[\hat{f}(x_0)])^2 = \mathbb{E}[(f(x_0) - \mathbb{E}[\hat{f}(x_0)])^2]\end{aligned}$$

So the equation could be written by

$$\begin{aligned}\mathbb{E}[(Y_0 - \hat{f}(x_0))^2] &= \mathbb{E}[(f(x_0) - \hat{f}(x_0))^2] + \text{var}(\epsilon) \\ &= \text{var}(\hat{f}(x_0)) + b_f(\hat{f}(x_0))^2 + \text{var}(\epsilon)\end{aligned}$$